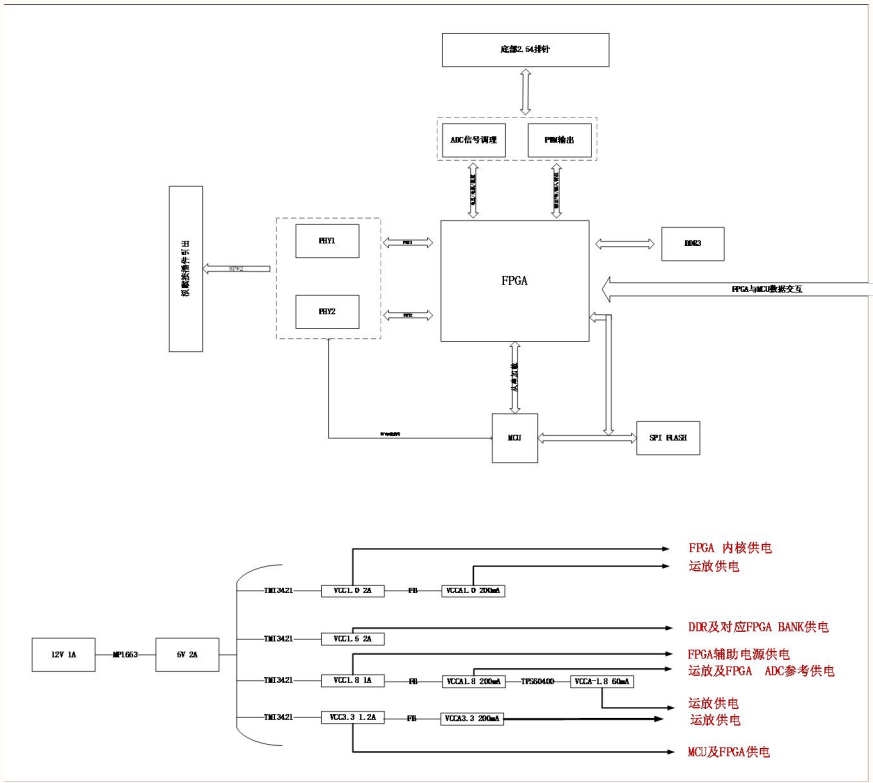
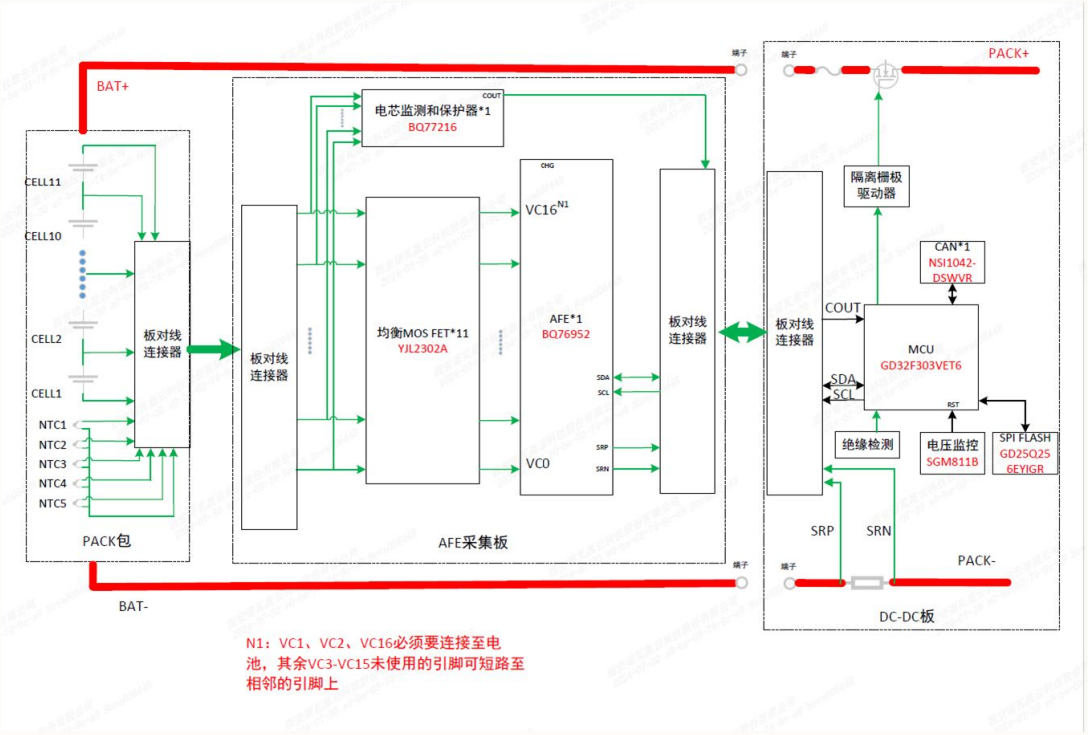



N1：VC1、VC2、VC16必须要连接至电池，其余VC3-VC15未使用的引脚可短路至相邻的引脚上

QR-CODE-500X500

MODEL

MESSAGE_BLOCK

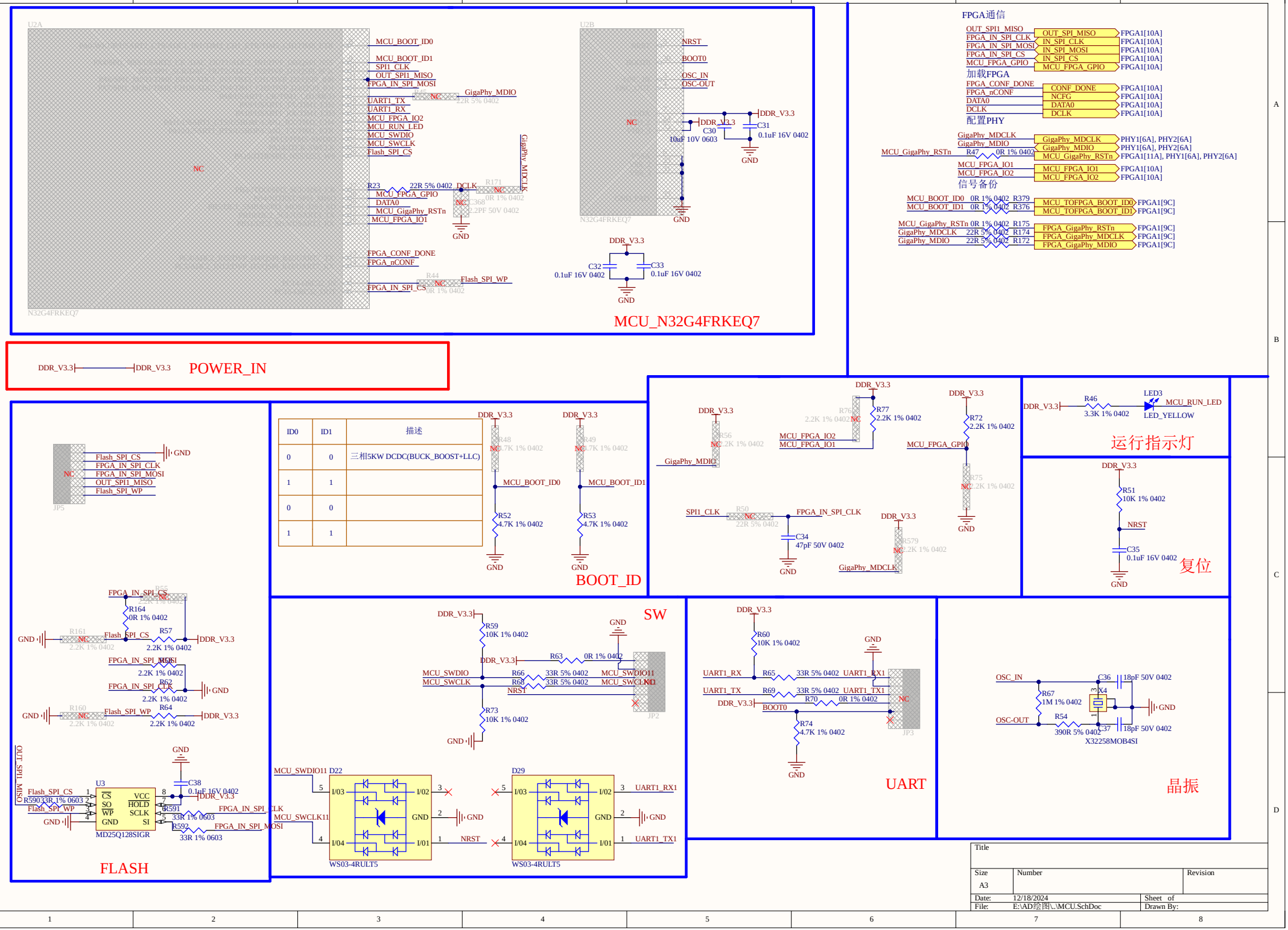
ESD_SILK

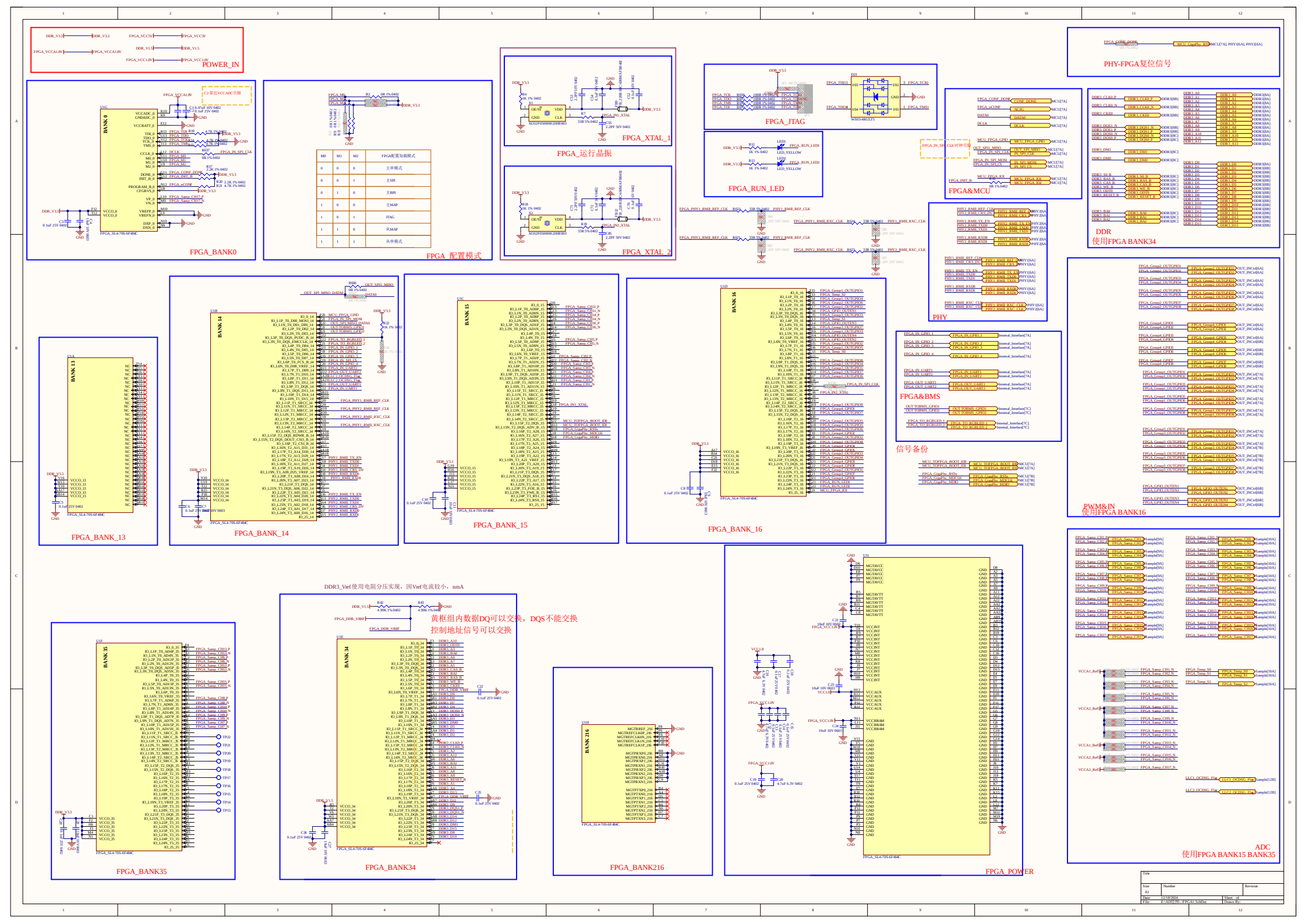
Date&QC

MARK2 Mark　MARK3 Mark　MARK6 Mark

MARK1 Mark　MARK4 Mark　MARK5 Mark

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| Title | | |
| Size A1 | Number | Revision |
| Date: 12/18/2024 | | Sheet of |
| File: E:\ADS\TC_TOP.SchDoc | | Drawn By: |


U2A
MCU_BOOT_ID0
MCU_BOOT_ID1
SPI1_CLK
OUT_SPI1_MISO
FPGA_IN_SPI_MOSI
GigaPhy_MDIO
UART1_TX
UART1_RX
MCU_FPGA_IO2
MCU_RUN_LED
MCU_SWDIO
MCU_SWCLK
Flash_SPI_CS
GigaPhy_MDCLK
R23 22R 5% 0402 DCLK
MCU_FPGA_GPIO
DATA0
MCU_GigaPhy_RSTn
MCU_FPGA_IO1
FPGA_CONF_DONE
FPGA_nCONF
Flash_SPI_WP
FPGA_IN_SPI_CS
N32G4FRKEQ7
U2B
NRST
BOOT0
OSC_IN
OSC-OUT
DDR_V3.3
C30 10uF 10V 0603
C31 0.1uF 16V 0402
C32 0.1uF 16V 0402
C33 0.1uF 16V 0402
GND
MCU_N32G4FRKEQ7
FPGA通信
OUT_SPI1_MISO
FPGA_IN_SPI_CLK
FPGA_IN_SPI_MOSI
FPGA_IN_SPI_CS
MCU_FPGA_GPIO
加载FPGA
FPGA_CONF_DONE
FPGA_nCONF
DATA0
DCLK
配置PHY
GigaPhy_MDCLK
GigaPhy_MDIO
MCU_GigaPhy_RSTn
R47 0R 1% 0402
信号备份
MCU_BOOT_ID0 0R 1% 0402 R379
MCU_BOOT_ID1 0R 1% 0402 R376
MCU_GigaPhy_RSTn 0R 1% 0402 R175
GigaPhy_MDCLK 22R 5% 0402 R174
GigaPhy_MDIO 22R 5% 0402 R172
FPGA1[10A]
PHY1[6A], PHY2[6A]
FPGA1[11A], PHY1[6A], PHY2[6A]
FPGA1[9C]
DDR_V3.3
DDR_V3.3
POWER_IN
JP5
Flash_SPI_CS
FPGA_IN_SPI_CLK
FPGA_IN_SPI_MOSI
OUT_SPI1_MISO
Flash_SPI_WP
GND
ID0
ID1
描述
三相5KW DCDC(BUCK_BOOST+LLC)
MCU_BOOT_ID0
MCU_BOOT_ID1
R52 4.7K 1% 0402
R53 4.7K 1% 0402
BOOT_ID
R56 2.2K 1% 0402
R76 2.2K 1% 0402
R77 2.2K 1% 0402
R72 2.2K 1% 0402
R75 2.2K 1% 0402
MCU_FPGA_IO2
MCU_FPGA_IO1
MCU_FPGA_GPIO
GigaPhy_MDIO
SPI1_CLK
FPGA_IN_SPI_CLK
C34 47pF 50V 0402
GigaPhy_MDCLK
R46 3.3K 1% 0402
LED3
LED_YELLOW
MCU_RUN_LED
运行指示灯
R51 10K 1% 0402
NRST
C35 0.1uF 16V 0402
复位
R164 0R 1% 0402
R57 2.2K 1% 0402
R62 2.2K 1% 0402
R64 2.2K 1% 0402
R161
R160
U3
MD25Q128SIGR
C38 0.1uF 16V 0402
R591 33R 1% 0603
R592 33R 1% 0603
FLASH
R59 10K 1% 0402
R63 0R 1% 0402
R66 33R 5% 0402
R68 33R 5% 0402
R73 10K 1% 0402
MCU_SWDIO
MCU_SWCLK
MCU_SWDIO11
MCU_SWCLK11
JP2
SW
D22
D29
WS03-4RULT5
R60 10K 1% 0402
R65 33R 5% 0402
R69 33R 5% 0402
R70 0R 1% 0402
R74 4.7K 1% 0402
UART1_RX
UART1_TX
UART1_RX1
UART1_TX1
JP3
UART
OSC_IN
OSC-OUT
C36 18pF 50V 0402
C37 18pF 50V 0402
R67 1M 1% 0402
R54 390R 5% 0402
X4
X32258MOB4SI
晶振
Title
Size A3
Number
Revision
Date: 12/18/2024
Sheet of
File: E:\AD绘图\..\MCU.SchDoc
Drawn By:


POWER_IN
FPGA_BANK0
FPGA_配置模式
FPGA_XTAL_1
FPGA_运行晶振
FPGA_XTAL_2
FPGA_JTAG
FPGA_RUN_LED
FPGA&MCU
PHY-FPGA复位信号
DDR
使用FPGA BANK34
PHY
FPGA&BMS
信号备份
PWM&IN
使用FPGA BANK16
FPGA_BANK_13
FPGA_BANK_14
FPGA_BANK_15
FPGA_BANK_16
DDR3_Vref使用电阻分压实现，因Vref电流较小，nmA
黄框组内数据DQ可以交换，DQS不能交换
控制地址信号可以交换
FPGA_BANK35
FPGA_BANK34
FPGA_BANK216
FPGA_POWER
ADC
使用FPGA BANK15 BANK35

## POWER_IN

DDR_V1.5 — DDR_V1.5

DDR_V3.3 — DDR_V3.3

## DDR3_MAIN

DDR3_Vref使用电阻分压实现，因Vref电流较小，nmA

FPGA_DDR_VREF

DDR_V1.5 — R156 4.99K 1% 0402 — R1574.99K 1% 0402 — GND

GND — C1390.1uF 16V 0402 — FPGA_DDR_VREF

C140 0.1uF 16V 0402

DDR_V1.5

IC3
M15T1G1664A-DEBG2S

M15T1G1664A-DEBG2CS

| | | | |
|---|---|---|---|
| DDR3_A0 | N3 | A0 | |
| DDR3_A1 | P7 | A1 | |
| DDR3_A2 | P3 | A2 | |
| DDR3_A3 | N2 | A3 | |
| DDR3_A4 | P8 | A4 | |
| DDR3_A5 | P2 | A5 | |
| DDR3_A6 | R8 | A6 | |
| DDR3_A7 | R2 | A7 | |
| DDR3_A8 | T8 | A8 | |
| DDR3_A9 | R3 | A9 | |
| DDR3_A10 | L7 | A10 | |
| DDR3_A11 | R7 | A11 | |
| DDR3_A12 | N7 | A12 | |
| DDR3_BA0 | M2 | BA0 | |
| DDR3_BA1 | N8 | BA1 | |
| DDR3_BA2 | M3 | BA2 | |
| DDR3_WE_B | L3 | WE | |
| DDR3_S0_B | L2 | CS | |
| DDR3_RAS_B | J3 | RAS | |
| DDR3_CAS_B | K3 | CAS | |
| DDR3_CLK0_P | J7 | CK | |
| DDR3_CLK0_N | K7 | CK | |
| DDR3_CKE0 | K9 | CKE | |
| DDR3_RESET_B | T2 | RESET | |
| DDR3_ODT0 | K1 | ODT | |
| | L8 | ZQ | |

| | | | |
|---|---|---|---|
| | | DQU0 | D7 DDR3_D8 |
| | | DQU1 | C3 DDR3_D9 |
| | | DQU2 | C8 DDR3_D10 |
| | | DQU3 | C2 DDR3_D11 |
| | | DQU4 | A7 DDR3_D12 |
| | | DQU5 | A2 DDR3_D13 |
| | | DQU6 | B8 DDR3_D14 |
| | | DQU7 | A3 DDR3_D15 |
| | | DQL0 | E3 DDR3_D0 |
| | | DQL1 | F7 DDR3_D1 |
| | | DQL2 | F2 DDR3_D2 |
| | | DQL3 | F8 DDR3_D3 |
| | | DQL4 | H3 DDR3_D4 |
| | | DQL5 | H8 DDR3_D5 |
| | | DQL6 | G2 DDR3_D6 |
| | | DQL7 | H7 DDR3_D7 |
| | | DQSU | B7 DDR3_DQS1_N |
| | | DQSU | C7 DDR3_DQS1_P |
| | | DQSL | F3 DDR3_DQS0_P |
| | | DQSL | G3 DDR3_DQS0_N |
| | | DMU | D3 DDR3_DM1 |
| | | DML | E7 DDR3_DM0 |
| | | NC1 | J1 |
| | | NC2 | J9 |
| | | NC3 | L1 |
| | | NC4 | L9 |
| | | NC5 | M7 |
| | | NC6 | T3 |
| | | NC7 | T7 |

R179 100R 1% 0402

GND — R185 240R 1% 0402

GND

C141 10uF 10V 0603 — C142 10uF 10V 0603 — C143 0.1uF 16V 0402 — C144 0.1uF 16V 0402 — C145 0.1uF 16V 0402 — C146 0.1uF 16V 0402 — C147 0.1uF 16V 0402 — C148 0.1uF 16V 0402 — C149 0.1uF 16V 0402 — C150 0.1uF 16V 0402

DDR_V1.5 — GND

## DDR_匹配电阻

R182 4.7K 1% 0402 — DDR3_S0_B
R183 4.7K 1% 0402 — DDR3_CKE0
R184 4.7K 1% 0402 — DDR3_RESET_B

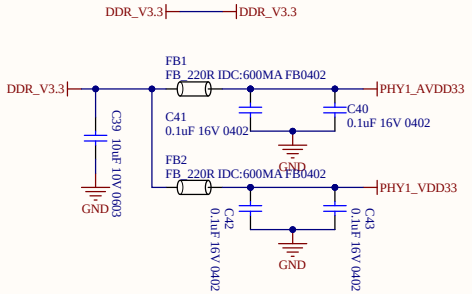
GND

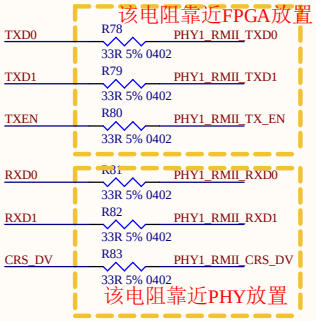
| | | |
|---|---|---|
| DDR_D | 数据总线 | 用于数据同步 |
| DQS | 数据选通 | 提供终端电阻，避免信号完整性问题 |
| DM | 数据屏蔽 | 写输入数据屏蔽信号 |
| DDR_D | 地址线 | 地址输入 |
| BA | BANK选通线 | 确定操作的具体BANK |
| CK/CK# | 差分时钟输入信号 | 控制及地址输入信号参考 |
| CS# | 使能脚 | 该信号为低时DDR使能 |
| CKE | 时钟使能输入引脚 | CKE为高有效 |
| RAS#\CAS# | 地址选通 | RAS行地址、CAS列地址选通脉冲 |
| ODT | 片上终端使能 | ODT使能/禁止片内终端电阻 |
| RESET | 复位信号 | 低电平有效、默认保持高电平 |
| ZQ | 阻抗匹配校准参考 | |
| | | |

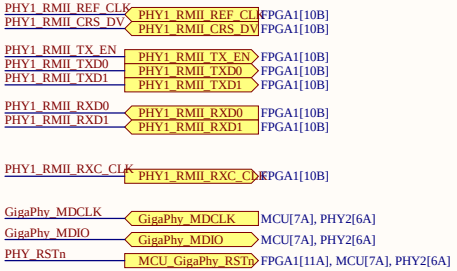
| | | |
|---|---|---|
| DDR3_A0 | DDR3_A0 | FPGA1[12A] |
| DDR3_A1 | DDR3_A1 | FPGA1[12A] |
| DDR3_A2 | DDR3_A2 | FPGA1[12A] |
| DDR3_A3 | DDR3_A3 | FPGA1[12A] |
| DDR3_A4 | DDR3_A4 | FPGA1[12A] |
| DDR3_A5 | DDR3_A5 | FPGA1[12A] |
| DDR3_A6 | DDR3_A6 | FPGA1[12A] |
| DDR3_A7 | DDR3_A7 | FPGA1[12A] |
| DDR3_A8 | DDR3_A8 | FPGA1[12A] |
| DDR3_A9 | DDR3_A9 | FPGA1[12A] |
| DDR3_A10 | DDR3_A10 | FPGA1[12A] |
| DDR3_A11 | DDR3_A11 | FPGA1[12A] |
| DDR3_A12 | DDR3_A12 | FPGA1[12A] |
| DDR3_D0 | DDR3_D0 | FPGA1[12A] |
| DDR3_D1 | DDR3_D1 | FPGA1[12A] |
| DDR3_D2 | DDR3_D2 | FPGA1[12A] |
| DDR3_D3 | DDR3_D3 | FPGA1[12A] |
| DDR3_D4 | DDR3_D4 | FPGA1[12A] |
| DDR3_D5 | DDR3_D5 | FPGA1[12A] |
| DDR3_D6 | DDR3_D6 | FPGA1[12A] |
| DDR3_D7 | DDR3_D7 | FPGA1[12A] |
| DDR3_D8 | DDR3_D8 | FPGA1[12A] |
| DDR3_D9 | DDR3_D9 | FPGA1[12A] |
| DDR3_D10 | DDR3_D10 | FPGA1[12A] |
| DDR3_D11 | DDR3_D11 | FPGA1[12A] |
| DDR3_D12 | DDR3_D12 | FPGA1[12A] |
| DDR3_D13 | DDR3_D13 | FPGA1[12A] |
| DDR3_D14 | DDR3_D14 | FPGA1[12A] |
| DDR3_D15 | DDR3_D15 | FPGA1[12A] |
| DDR3_CLK0_P | DDR3_CLK0_P | FPGA1[11A] |
| DDR3_CLK0_N | DDR3_CLK0_N | FPGA1[11A] |
| DDR3_CKE0 | DDR3_CKE0 | FPGA1[11A] |
| DDR3_DQS1_N | DDR3_DQS1_N | FPGA1[11A] |
| DDR3_DQS1_P | DDR3_DQS1_P | FPGA1[11A] |
| DDR3_DQS0_N | DDR3_DQS0_N | FPGA1[11A] |
| DDR3_DQS0_P | DDR3_DQS0_P | FPGA1[11A] |
| DDR3_DM1 | DDR3_DM1 | FPGA1[11A] |
| DDR3_DM0 | DDR3_DM0 | FPGA1[11A] |
| DDR3_S0_B | DDR3_S0_B | FPGA1[11A] |
| DDR3_RAS_B | DDR3_RAS_B | FPGA1[11A] |
| DDR3_CAS_B | DDR3_CAS_B | FPGA1[11A] |
| DDR3_WE_B | DDR3_WE_B | FPGA1[11A] |
| DDR3_ODT0 | DDR3_ODT0 | FPGA1[11A] |
| DDR3_RESET_B | DDR3_RESET_B | FPGA1[11A] |
| DDR3_BA0 | DDR3_BA0 | FPGA1[11A] |
| DDR3_BA1 | DDR3_BA1 | FPGA1[11A] |
| DDR3_BA2 | DDR3_BA2 | FPGA1[11A] |

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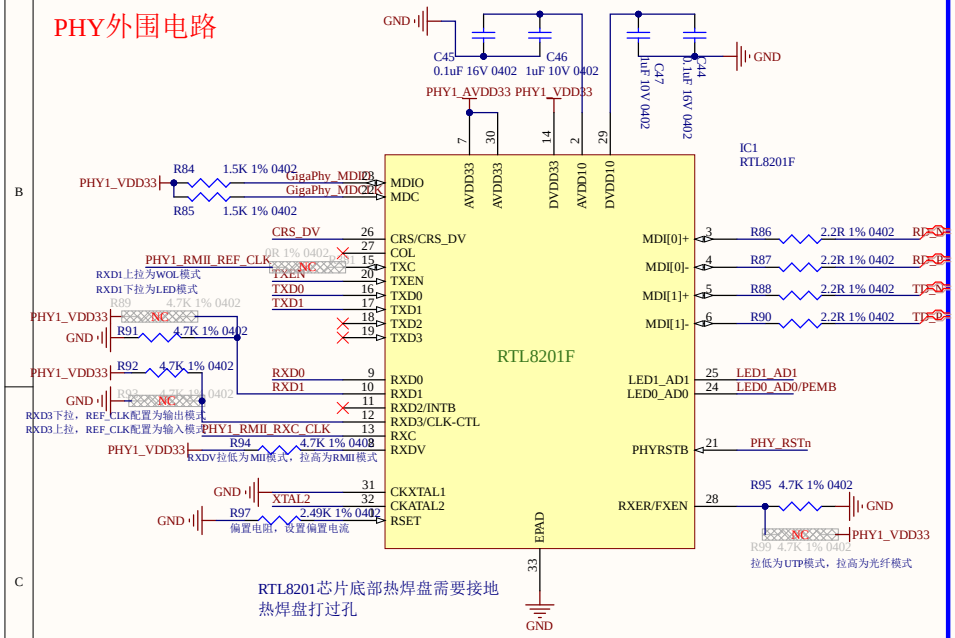
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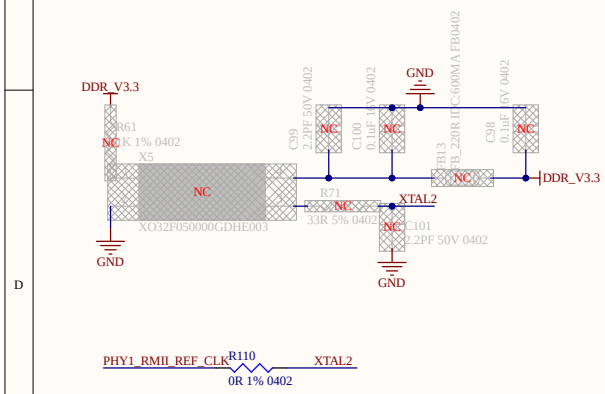



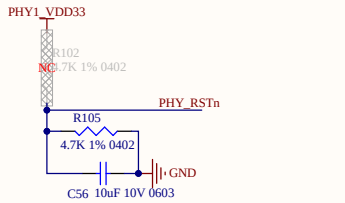

## PHY外围电路



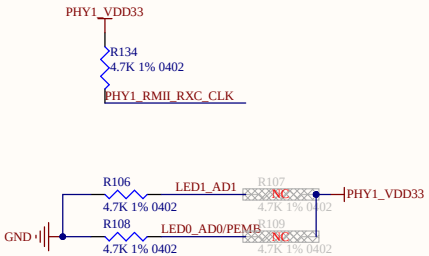

RTL8201芯片底部热焊盘需要接地
热焊盘打过孔




焊接RTL8201时，焊接R127,晶振电路NP，焊接YT8512取反

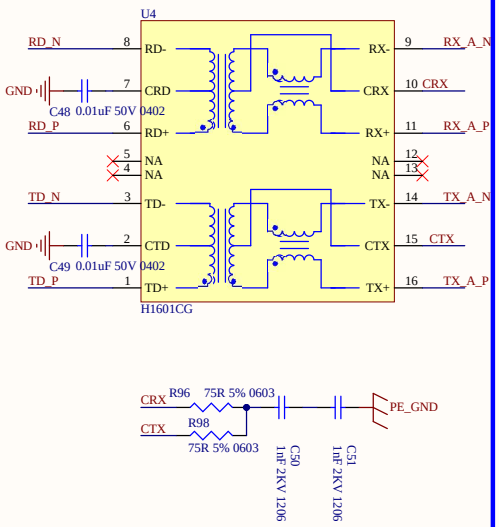



Note：1、在PHY的3.3V稳定之前复位为高至少80us；
2、要PHY完全复位，至少拉低需要10ms；
3、在PHY复位为高时至少150ms之后才可以访问PHY的寄存器。
4、复位与主板上PHY复位复用，因此，此处RC NP

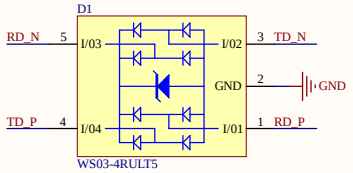



PHY地址配置为00

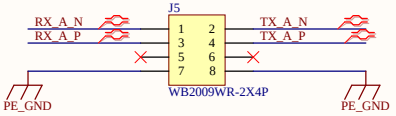
## 网络变压器




## TVS

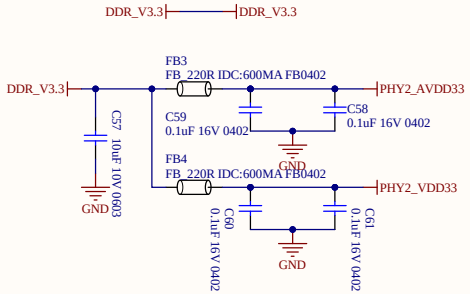


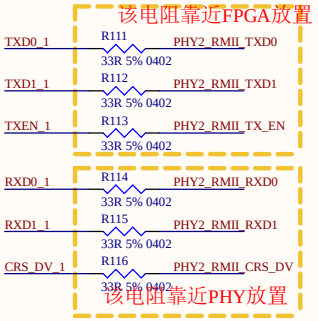

## 端子引出

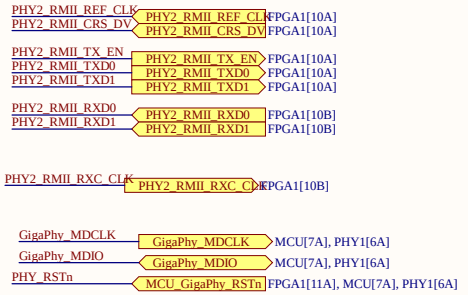



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| Date: 12/18/2024 | | Sheet of |
| File: E:\AD绘图\..\PHY1.SchDoc | | Drawn By: |

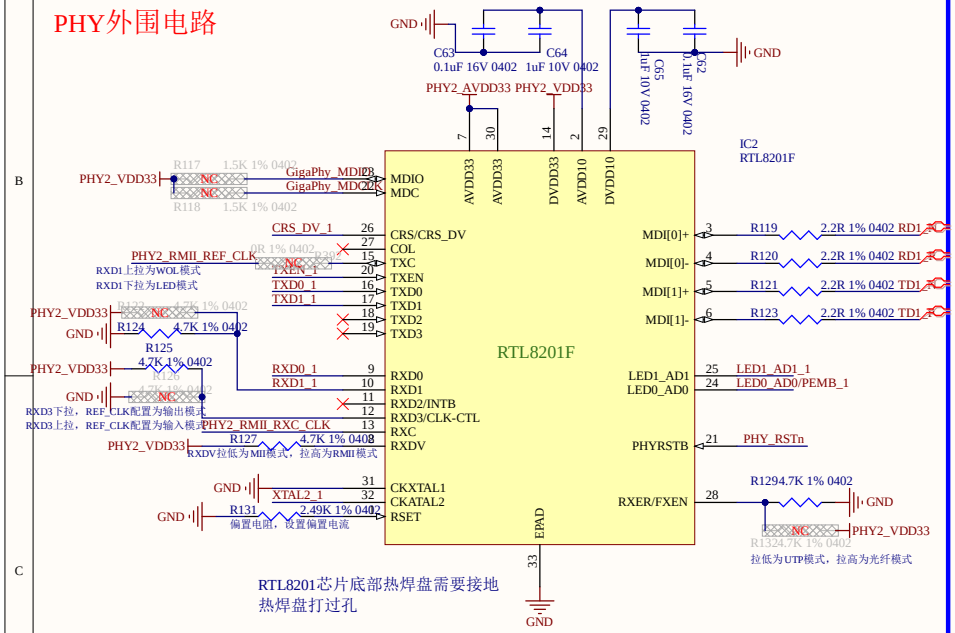
## Power



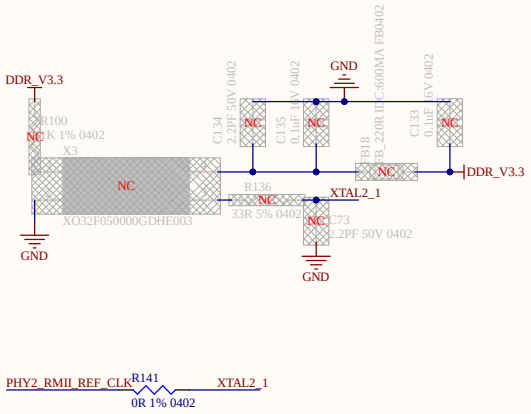




## PHY外围电路

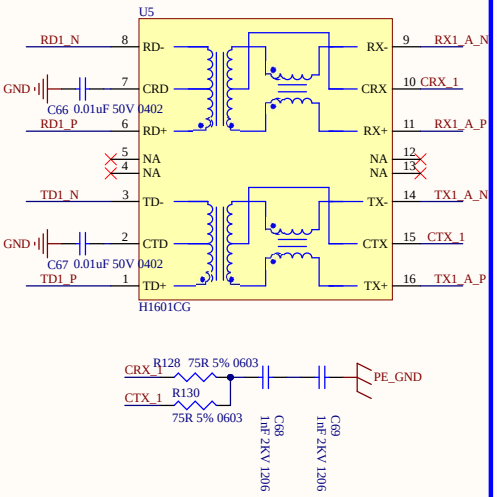



RTL8201芯片底部热焊盘需要接地
热焊盘打过孔



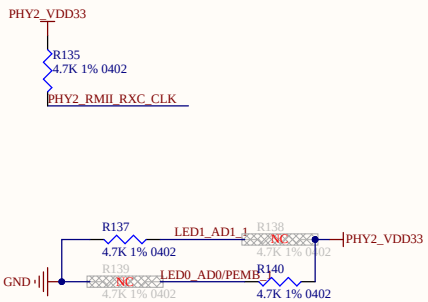

焊接RTL8201时，焊接R127,晶振电路NP，焊接YT8512取反

## 网络变压器



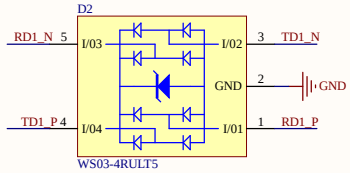




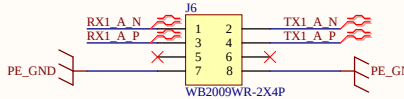



PHY地址配置为01

## TVS




## 端子引出

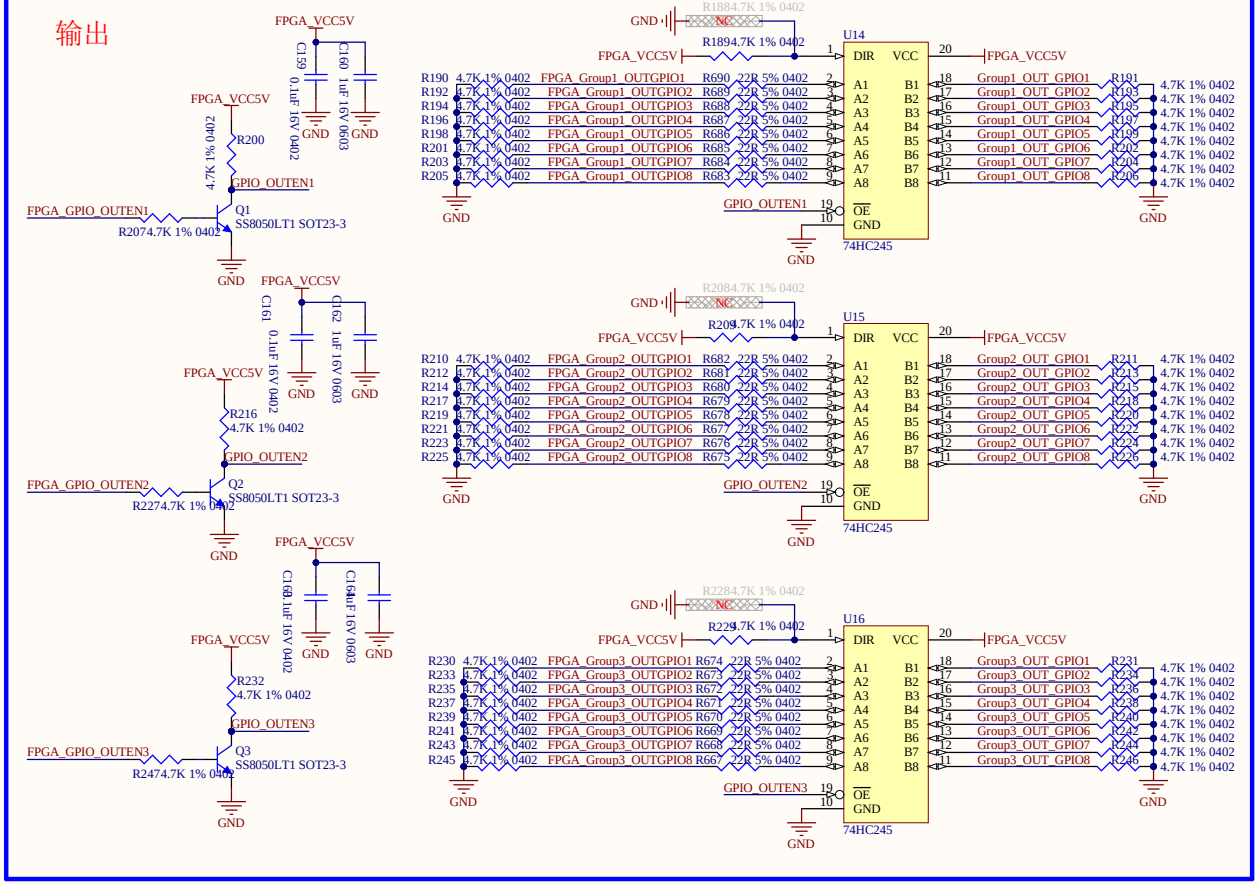



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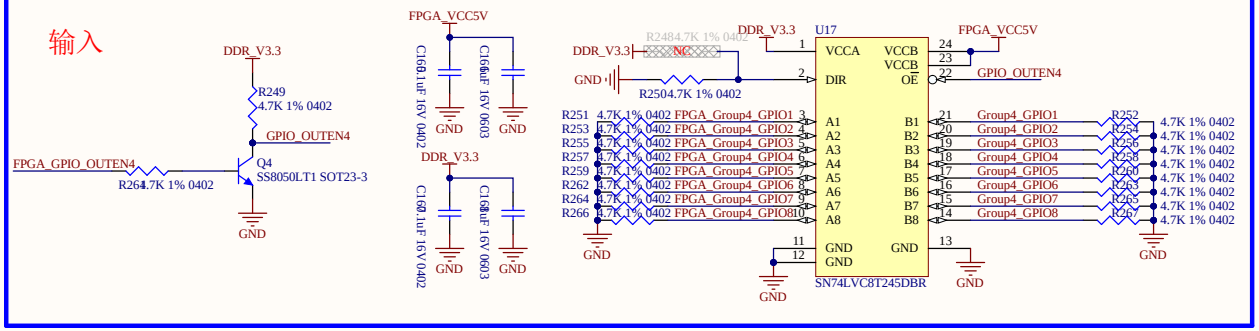
FPGA_VCC5V — FPGA_VCC5V
DDR_V3.3 — DDR_V3.3

# POWER_IN

## 输出




## 输入




拉高使能PWM输出，拉低之后PWM为0
245之前下拉是为了加载程序的时候防止FPGA输出不定态
245之后下拉是为了刹车信号来的时候，给到驱动板的是低电平

FPGA_Group2_OUTGPIO1 — FPGA_Group2_OUTGPIO1 — FPGA1[11B]
FPGA_Group2_OUTGPIO2 — FPGA_Group2_OUTGPIO2 — FPGA1[11B]
FPGA_Group2_OUTGPIO3 — FPGA_Group2_OUTGPIO3 — FPGA1[11B]
FPGA_Group2_OUTGPIO4 — FPGA_Group2_OUTGPIO4 — FPGA1[11B]
FPGA_Group2_OUTGPIO5 — FPGA_Group2_OUTGPIO5 — FPGA1[11B]
FPGA_Group2_OUTGPIO6 — FPGA_Group2_OUTGPIO6 — FPGA1[11B]
FPGA_Group2_OUTGPIO7 — FPGA_Group2_OUTGPIO7 — FPGA1[11B]
FPGA_Group2_OUTGPIO8 — FPGA_Group2_OUTGPIO8 — FPGA1[11B]

FPGA_Group1_OUTGPIO1 — FPGA_Group1_OUTGPIO1 — FPGA1[11B]
FPGA_Group1_OUTGPIO2 — FPGA_Group1_OUTGPIO2 — FPGA1[11B]
FPGA_Group1_OUTGPIO3 — FPGA_Group1_OUTGPIO3 — FPGA1[11B]
FPGA_Group1_OUTGPIO4 — FPGA_Group1_OUTGPIO4 — FPGA1[11B]
FPGA_Group1_OUTGPIO5 — FPGA_Group1_OUTGPIO5 — FPGA1[11B]
FPGA_Group1_OUTGPIO6 — FPGA_Group1_OUTGPIO6 — FPGA1[11B]
FPGA_Group1_OUTGPIO7 — FPGA_Group1_OUTGPIO7 — FPGA1[11B]
FPGA_Group1_OUTGPIO8 — FPGA_Group1_OUTGPIO8 — FPGA1[11B]

FPGA_Group4_GPIO1 — FPGA_Group4_GPIO1 — FPGA1[11B]
FPGA_Group4_GPIO2 — FPGA_Group4_GPIO2 — FPGA1[11B]
FPGA_Group4_GPIO3 — FPGA_Group4_GPIO3 — FPGA1[11B]
FPGA_Group4_GPIO4 — FPGA_Group4_GPIO4 — FPGA1[11B]
FPGA_Group4_GPIO5 — FPGA_Group4_GPIO5 — FPGA1[11B]
FPGA_Group4_GPIO6 — FPGA_Group4_GPIO6 — FPGA1[11B]
FPGA_Group4_GPIO7 — FPGA_Group4_GPIO7 — FPGA1[11B]
FPGA_Group4_GPIO8 — FPGA_Group4_GPIO8 — FPGA1[11B]

FPGA_Group3_OUTGPIO1 — FPGA_Group3_OUTGPIO1 — FPGA1[11B]
FPGA_Group3_OUTGPIO2 — FPGA_Group3_OUTGPIO2 — FPGA1[11B]
FPGA_Group3_OUTGPIO3 — FPGA_Group3_OUTGPIO3 — FPGA1[11B]
FPGA_Group3_OUTGPIO4 — FPGA_Group3_OUTGPIO4 — FPGA1[11B]
FPGA_Group3_OUTGPIO5 — FPGA_Group3_OUTGPIO5 — FPGA1[11B]
FPGA_Group3_OUTGPIO6 — FPGA_Group3_OUTGPIO6 — FPGA1[11B]
FPGA_Group3_OUTGPIO7 — FPGA_Group3_OUTGPIO7 — FPGA1[11C]
FPGA_Group3_OUTGPIO8 — FPGA_Group3_OUTGPIO8 — FPGA1[11C]

FPGA_GPIO_OUTEN1 — FPGA_GPIO_OUTEN1 — FPGA1[11C]
FPGA_GPIO_OUTEN2 — FPGA_GPIO_OUTEN2 — FPGA1[11C]
FPGA_GPIO_OUTEN3 — FPGA_GPIO_OUTEN3 — FPGA1[11C]
FPGA_GPIO_OUTEN4 — FPGA_GPIO_OUTEN4 — FPGA1[11C]

Group2_OUT_GPIO1 — Group2_OUTGPIO1 — Interface[6B] — Group1_OUT_GPIO1 — Group1_OUTGPIO1 — Intemal_Interface[7A]
Group2_OUT_GPIO2 — Group2_OUTGPIO2 — Interface[6B] — Group1_OUT_GPIO2 — Group1_OUTGPIO2 — Intemal_Interface[7A]
Group2_OUT_GPIO3 — Group2_OUTGPIO3 — Interface[6B] — Group1_OUT_GPIO3 — Group1_OUTGPIO3 — Intemal_Interface[7A]
Group2_OUT_GPIO4 — Group2_OUTGPIO4 — Interface[6B] — Group1_OUT_GPIO4 — Group1_OUTGPIO4 — Intemal_Interface[7A]
Group2_OUT_GPIO5 — Group2_OUTGPIO5 — Interface[6B] — Group1_OUT_GPIO5 — Group1_OUTGPIO5 — Interface[7B]
Group2_OUT_GPIO6 — Group2_OUTGPIO6 — Interface[6B] — Group1_OUT_GPIO6 — Group1_OUTGPIO6 — Interface[7B]
Group2_OUT_GPIO7 — Group2_OUTGPIO7 — Interface[6B] — Group1_OUT_GPIO7 — Group1_OUTGPIO7 — Interface[7B]
Group2_OUT_GPIO8 — Group2_OUTGPIO8 — Interface[6B] — Group1_OUT_GPIO8 — Group1_OUTGPIO8 — Interface[7B]

Group4_GPIO1 — Group4_GPIO1 — Interface[6B] — Group3_OUT_GPIO1 — Group3_OUTGPIO1 — Interface[7C]
Group4_GPIO2 — Group4_GPIO2 — Interface[6B] — Group3_OUT_GPIO2 — Group3_OUTGPIO2 — Interface[7C]
Group4_GPIO3 — Group4_GPIO3 — Interface[6B] — Group3_OUT_GPIO3 — Group3_OUTGPIO3 — Interface[7C]
Group4_GPIO4 — Group4_GPIO4 — Interface[6B] — Group3_OUT_GPIO4 — Group3_OUTGPIO4 — Interface[7C]
Group4_GPIO5 — Group4_GPIO5 — Interface[6C] — Group3_OUT_GPIO5 — Group3_OUTGPIO5 — Interface[7C]
Group4_GPIO6 — Group4_GPIO6 — Interface[6C] — Group3_OUT_GPIO6 — Group3_OUTGPIO6 — Interface[7C]
Group4_GPIO7 — Group4_GPIO7 — Interface[6C] — Group3_OUT_GPIO7 — Group3_OUTGPIO7 — Interface[7C]
Group4_GPIO8 — Group4_GPIO8 — Interface[6C] — Group3_OUT_GPIO8 — Group3_OUTGPIO8 — Interface[7C]

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输入14V，对应采样0.95V，采样带宽4K

电压采样

电流值 电压值 采样输出 输入1.3V，对应采样0.959V，采样带宽60K

16.25A 1.3V 0.958V
10A 0.8V 0.782V
5A 0.4V 0.641V
Vo=0.3529Vi+0.5

电流采样

Vref=0.503

基准电压0.868V

基准电压0.868V

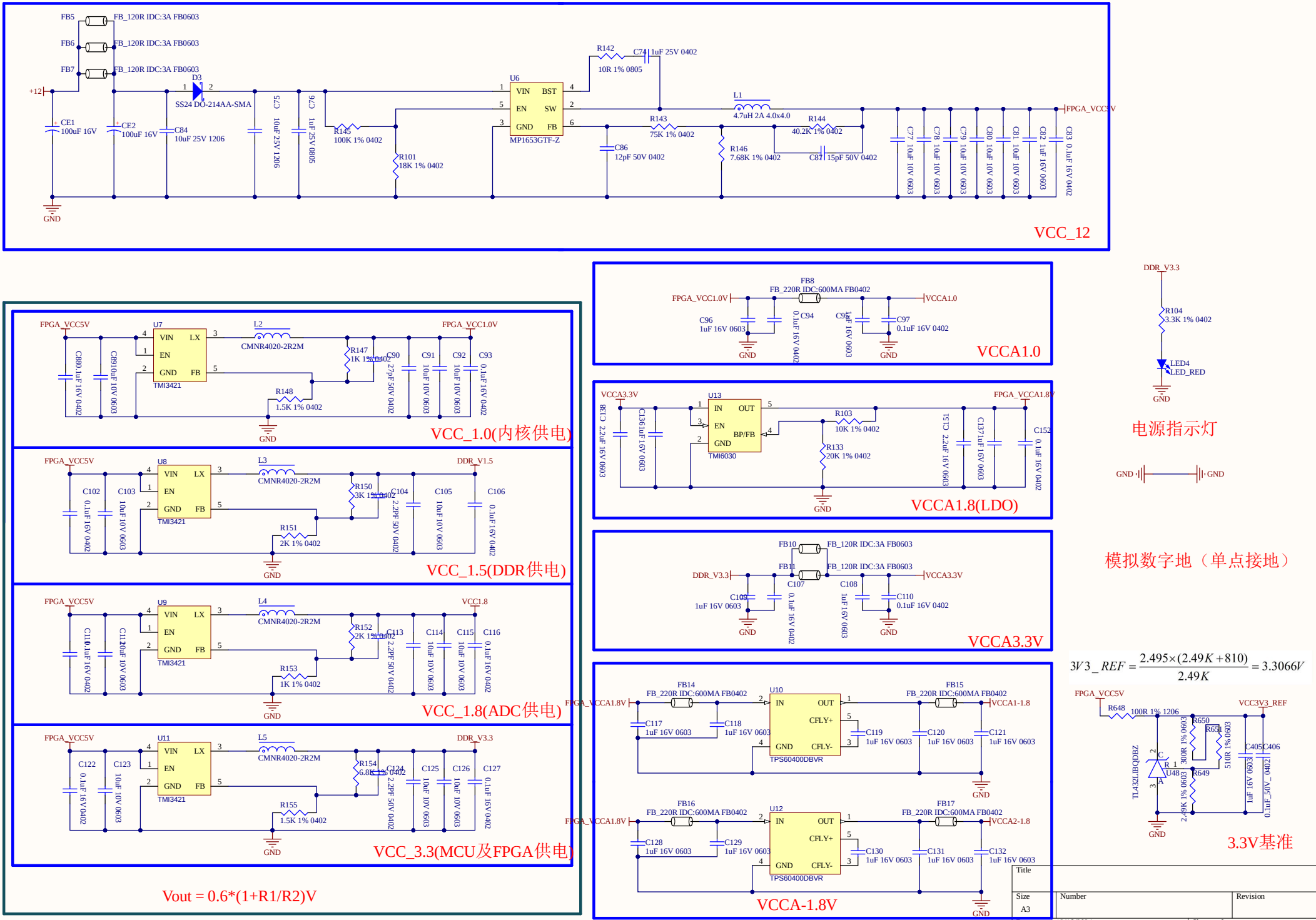
过流信号

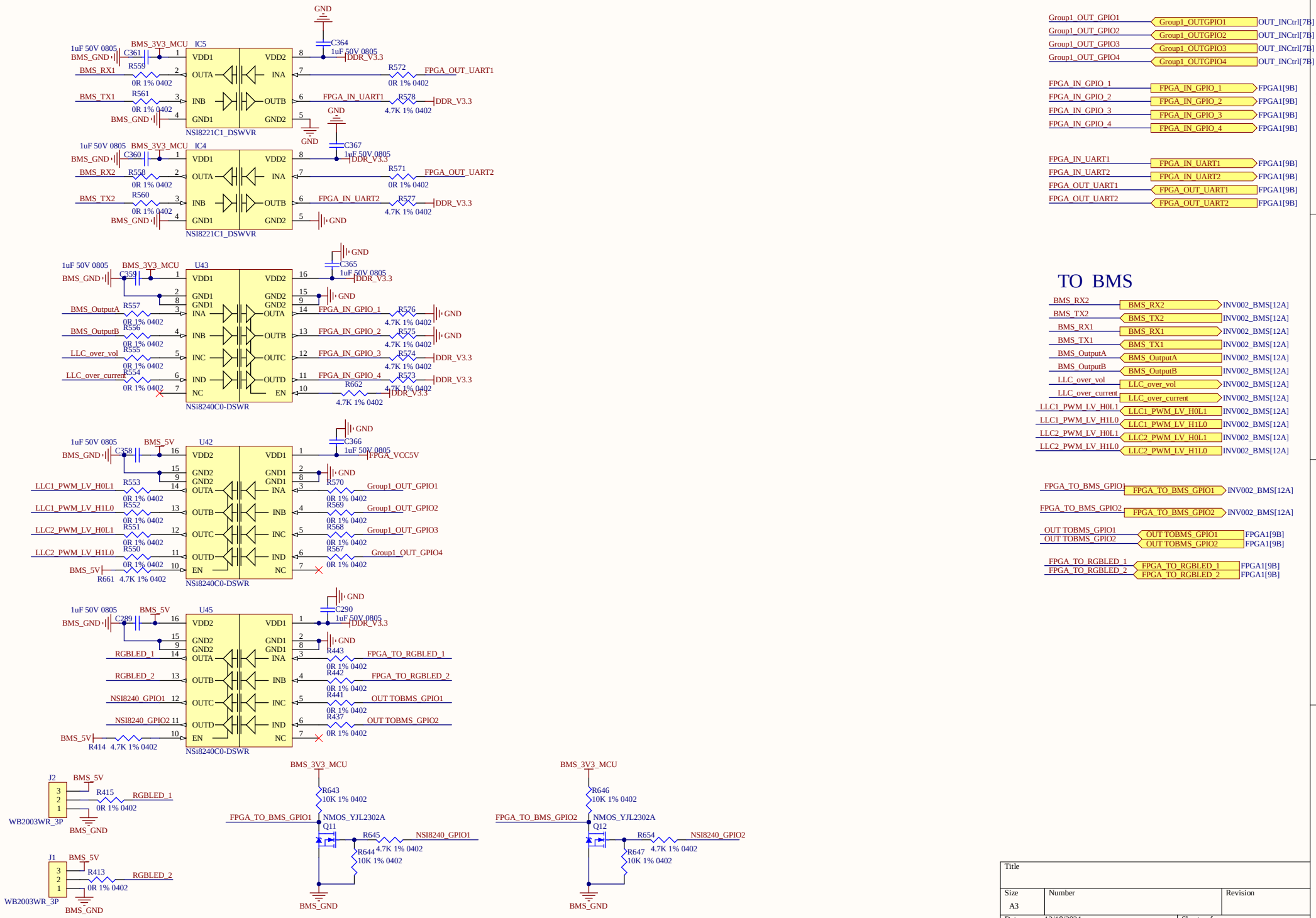
LLC谐振腔硬件过流保护

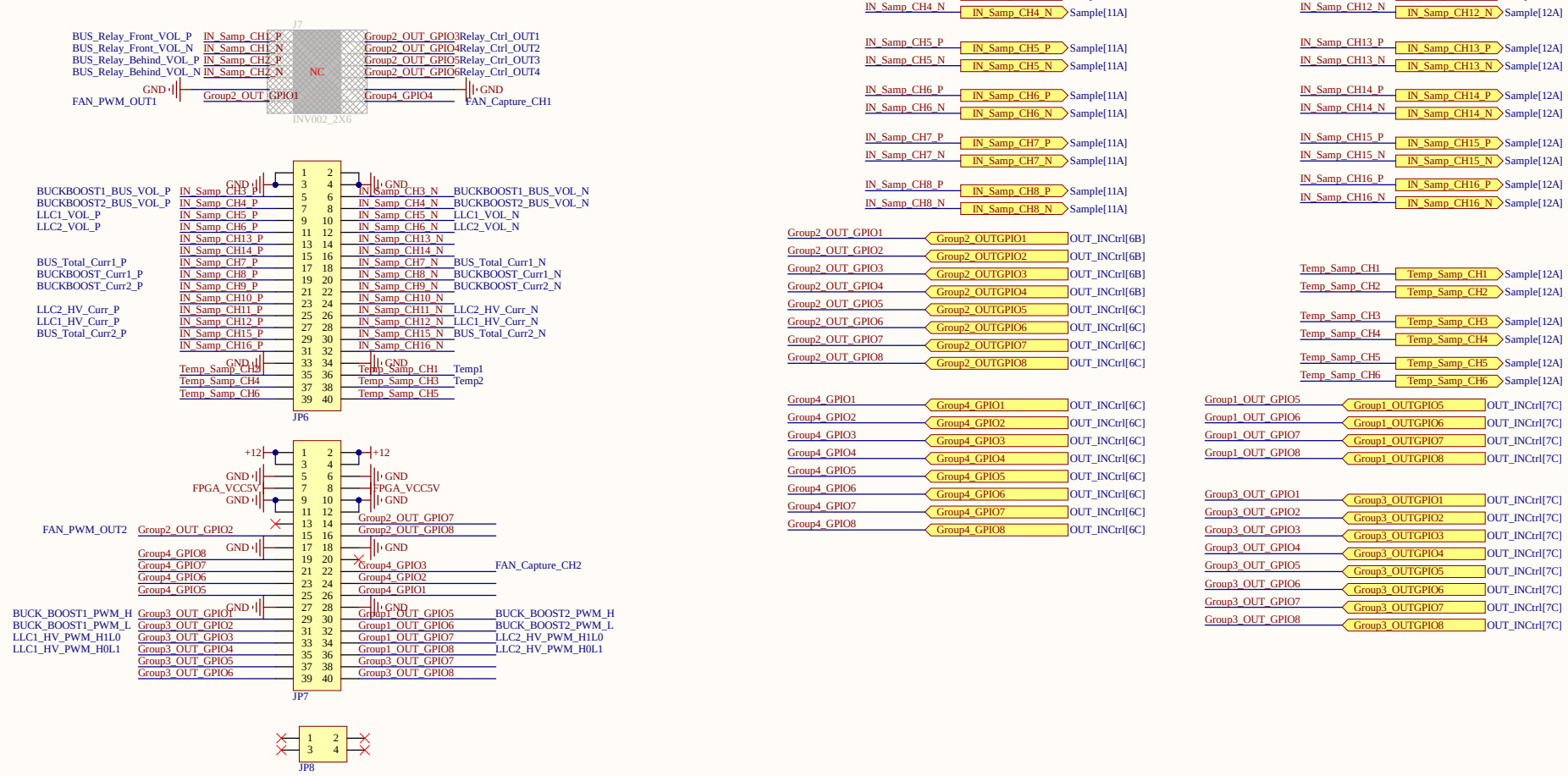
输入3.3V，对应采样780mV，采样带宽40K

ADC_17

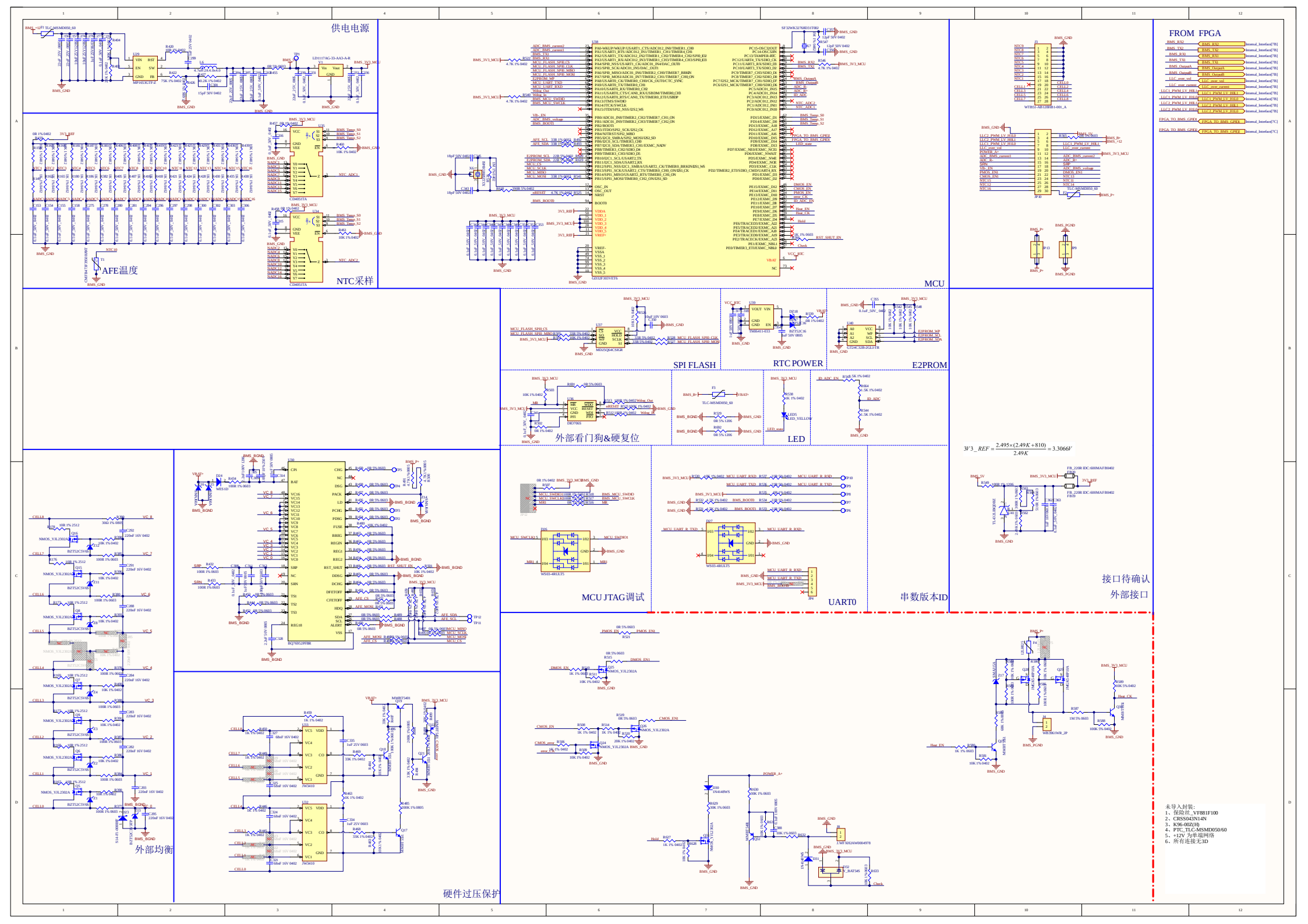
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FB5 FB_120R IDC:3A FB0603
FB6 FB_120R IDC:3A FB0603
FB7 FB_120R IDC:3A FB0603
+12
D3 SS24 DO-214AA-SMA
CE1 100uF 16V
CE2 100uF 16V
C84 10uF 25V 1206
C77 10uF 25V 1206
C79 1uF 25V 0805
R145 100K 1% 0402
R101 18K 1% 0402
U6 MP1653GTF-Z
VIN BST EN SW GND FB
R142 10R 1% 0805
C74 1uF 25V 0402
L1 4.7uH 2A 4.0x4.0
R143 75K 1% 0402
C86 12pF 50V 0402
R146 7.68K 1% 0402
R144 40.2K 1% 0402
C87 15pF 50V 0402
C77 10uF 10V 0603
C78 10uF 10V 0603
C79 10uF 10V 0603
C80 10uF 10V 0603
C81 10uF 10V 0603
C82 1uF 16V 0603
C83 0.1uF 16V 0402
FPGA_VCC12V
GND
VCC_12
FPGA_VCC5V
U7 TMI3421
VIN LX EN GND FB
L2 CMNR4020-2R2M
FPGA_VCC1.0V
C88 1uF 16V 0402
C89 10uF 10V 0603
R147 1K 1% 0402
R148 1.5K 1% 0402
C90 27pF 50V 0402
C91 10uF 10V 0603
C92 10uF 10V 0603
C93 0.1uF 16V 0402
VCC_1.0(内核供电)
U8 TMI3421
L3 CMNR4020-2R2M
DDR_V1.5
C102 0.1uF 16V 0402
C103 10uF 10V 0603
R150 3K 1% 0402
R151 2K 1% 0402
C104 2.2PF 50V 0402
C105 10uF 10V 0603
C106 0.1uF 16V 0402
VCC_1.5(DDR供电)
U9 TMI3421
L4 CMNR4020-2R2M
VCC1.8
C110 0.1uF 16V 0402
C112 10uF 10V 0603
R152 2K 1% 0402
R153 1K 1% 0402
C113 2.2PF 50V 0402
C114 10uF 10V 0603
C115 10uF 10V 0603
C116 0.1uF 16V 0402
VCC_1.8(ADC供电)
U11 TMI3421
L5 CMNR4020-2R2M
DDR_V3.3
C122 0.1uF 16V 0402
C123 10uF 10V 0603
R154 6.8K 1% 0402
R155 1.5K 1% 0402
C124 2.2PF 50V 0402
C125 10uF 10V 0603
C126 10uF 10V 0603
C127 0.1uF 16V 0402
VCC_3.3(MCU及FPGA供电)
Vout = 0.6*(1+R1/R2)V
FB8 FB_220R IDC:600MA FB0402
FPGA_VCC1.0V
VCCA1.0
C96 1uF 16V 0603
C94 0.1uF 16V 0402
C95 1uF 16V 0603
C97 0.1uF 16V 0402
VCCA1.0
VCCA3.3V
U13 TMI6030
IN OUT EN BP/FB GND
FPGA_VCCA1.8V
C138 2.2uF 16V 0603
C136 1uF 16V 0603
R103 10K 1% 0402
R133 20K 1% 0402
C151 2.2uF 16V 0603
C137 1uF 16V 0603
C152 0.1uF 16V 0402
VCCA1.8(LDO)
FB10 FB_120R IDC:3A FB0603
FB11 FB_120R IDC:3A FB0603
DDR_V3.3
VCCA3.3V
C107
C109 1uF 16V 0603
C108 1uF 16V 0603
C110 0.1uF 16V 0402
VCCA3.3V
FB14 FB_220R IDC:600MA FB0402
FB15 FB_220R IDC:600MA FB0402
FB16 FB_220R IDC:600MA FB0402
FB17 FB_220R IDC:600MA FB0402
U10 U12 TPS60400DBVR
IN OUT CFLY+ GND CFLY-
FPGA_VCCA1.8V
VCCA1-1.8
VCCA2-1.8
C117 1uF 16V 0603
C118 1uF 16V 0603
C119 1uF 16V 0603
C120 1uF 16V 0603
C121 1uF 16V 0603
C128 1uF 16V 0603
C129 1uF 16V 0603
C130 1uF 16V 0603
C131 1uF 16V 0603
C132 1uF 16V 0603
VCCA-1.8V
DDR_V3.3
R104 3.3K 1% 0402
LED4 LED_RED
电源指示灯
GND
模拟数字地（单点接地）
$3V3_REF = \frac{2.495\times(2.49K+810)}{2.49K} = 3.3066V$
FPGA_VCC5V
R648 100R 1% 1206
VCC3V3_REF
R650 300R 1% 0603
R651 510R 1% 0603
U48 TL432LIBQDBZ
R649 2.49K 1% 0603
C405 1uF 16V 0603
C406 0.1uF 50V 0402
3.3V基准
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GND
BMS_3V3_MCU
IC5
1uF 50V 0805
BMS_GND
C361
R559
BMS_RX1
0R 1% 0402
R561
BMS_TX1
BMS_GND
VDD1
OUTA
INB
GND1
VDD2
INA
OUTB
GND2
C364
1uF 50V 0805
DDR_V3.3
R572
FPGA_OUT_UART1
FPGA_IN_UART1
R578
4.7K 1% 0402
NSI8221C1_DSWVR
C367
R571
FPGA_OUT_UART2
R560
BMS_TX2
R577
IC4
C360
R558
BMS_RX2
U43
C359
BMS_OutputA
R557
BMS_OutputB
R556
LLC_over_vol
R555
LLC_over_current
R554
GND1
INC
IND
NC
GND2
OUTC
OUTD
EN
C365
R576
FPGA_IN_GPIO_1
R575
FPGA_IN_GPIO_2
R574
FPGA_IN_GPIO_3
R573
FPGA_IN_GPIO_4
R662
4.7K 1% 0402
NSI8240C0-DSWR
U42
C358
BMS_5V
R661
C366
FPGA_VCC5V
R553
LLC1_PWM_LV_H0L1
R552
LLC1_PWM_LV_H1L0
R551
LLC2_PWM_LV_H0L1
R550
LLC2_PWM_LV_H1L0
R570
Group1_OUT_GPIO1
R569
Group1_OUT_GPIO2
R568
Group1_OUT_GPIO3
R567
Group1_OUT_GPIO4
U45
C289
C290
RGBLED_1
RGBLED_2
NSI8240_GPIO1
NSI8240_GPIO2
R443
FPGA_TO_RGBLED_1
R442
FPGA_TO_RGBLED_2
R441
OUT TOBMS_GPIO1
R437
OUT TOBMS_GPIO2
R414
J2
R415
WB2003WR_3P
J1
R413
R643
10K 1% 0402
FPGA_TO_BMS_GPIO1
NMOS_YJL2302A
Q11
R645
R644
4.7K 1% 0402
R646
FPGA_TO_BMS_GPIO2
Q12
R654
R647
Group1_OUTGPIO1
Group1_OUTGPIO2
Group1_OUTGPIO3
Group1_OUTGPIO4
OUT_INCtrl[7B]
FPGA_IN_GPIO_1
FPGA_IN_GPIO_2
FPGA_IN_GPIO_3
FPGA_IN_GPIO_4
FPGA1[9B]
FPGA_IN_UART1
FPGA_IN_UART2
FPGA_OUT_UART1
FPGA_OUT_UART2
TO BMS
INV002_BMS[12A]
FPGA_TO_BMS_GPIO1
FPGA_TO_BMS_GPIO2
OUT TOBMS_GPIO1
OUT TOBMS_GPIO2
FPGA_TO_RGBLED_1
FPGA_TO_RGBLED_2
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供电电源
AFE温度
NTC采样
MCU
FROM FPGA
SPI FLASH
RTC POWER
E2PROM
外部看门狗&硬复位
LED
MCU JTAG调试
UART0
串数版本ID
接口待确认
外部接口
外部均衡
硬件过压保护
未导入封装:
1、保险丝_VFB81F100
2、CRSS043N14N
3、K96-002(H)
4、PTC_TLC-MSMD050/60
5、+12V 为单端网络
6、所有连接无3D